SLEEVEDRIFT · REGULATORY AND STANDARDS REGISTER

SleeveDrift regulatory and standards register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-02	Amish Chadha	First issue at the kit 1.9.1 upgrade; confined-space, rope, gas, machinery and risk standards that touch a rescue kit; not legal advice

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SleeveDrift is published as an open engineering reference, not as certified rescue, mining or confined-space equipment, so no certification is needed to publish the design. The register lists what a team that builds, stages or uses a kit would meet, so the design can be shaped for it. The rules on the people who enter a pipe (permits, rescue plans, training) fall on the operator, not on the design. Not legal advice; anything marked "to assess" needs a qualified reviewer, and the use of the kit in a specific country needs the local authority's view before any claim.

> **Safety:** Digging by hand in a rescue pipe is confined-space work. Meeting a row below on paper does not make the kit safe to use. Nothing here replaces an incident commander, a standby crew at the portal and continuous gas monitoring.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	29 CFR 1910.146, Permit-required confined spaces (OSHA)	United States	Permit programme, atmospheric testing, attendants, rescue and retrieval for entry into confined spaces	The pipe under collapse debris is a confined space. The kit supplies retrieval (casualty line, stretcher), air and gas monitoring that a permit programme would require; the permit and the rescue service are the operator's	Operator and incident commander	Applies to the operator; the kit supports it (SVD-CAL-001 E, F; PR-006, PR-005, PR-012)
G2	29 CFR 1926 Subpart AA, Confined spaces in construction (OSHA)	United States	The same duties for construction sites, which include tunnel works	Tunnel contractors who hold a kit would work under it. Section numbers and the retrieval rules to read against the kit	Tunnel contractor	To assess
G3	NFPA 1670, Technical rescue operations (NFPA)	International practice, US origin	Levels of rescue capability, including confined-space rescue	Sets what a rescue team should be trained for before it uses a kit in a pipe; no equipment claim	Rescue commander	To assess
G4	Gas detection: portable four-gas monitors for oxygen, carbon monoxide, hydrogen sulphide and flammable gas	International practice	Atmospheric monitoring in confined spaces	The kit carries two bought monitors (BOM line 25); an example product is the Honeywell BW GasAlertMicroClip XL, GBP 776.52 excluding VAT, checked 2026-11-02 (Sources). Bump testing before each shift is a use rule	Rescue commander	Met by design (PR-012); the monitor choice, calibration and the area classification of a collapse atmosphere to assess
G5	ISO 12100:2010 (EN ISO 12100:2010), Safety of machinery, General principles for design, Risk assessment and risk reduction	International; Europe	The method for hazard analysis and risk reduction on machinery	The portal winches with drill drive are machinery in the sense of the standard. SVD-RSK-001 follows its hazard method in a preliminary form	Design lead	Applies as method; not a certification
G6	NFPA 1983 (consolidated into NFPA 2500 on 2021-09-15), Life safety rope and equipment	United States	Design, test and labelling of rope and hardware for life-safety use, with a 15:1 rope factor	The pull rope also carries a casualty, so life-safety practice is the benchmark. 15 x the 519 N starting pull is 7.8 kN against 16.2 kN of rope after the splice (STEM S17). The bought	Procurement	To assess (rope marking); the factor is met on paper

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
	for emergency services			10 mm polyester rope is not marked to the standard; choosing a marked rope is a to-assess item		
G7	Directive 2006/42/EC, Machinery, and Regulation (EU) 2023/1230, which replaces it and applies from 2027-01-20	European Union	Essential health and safety requirements, CE marking for machinery put on the market	Applies only to someone who sells a made kit in the EU. The open design is not a product on the market. The drill-driven winch would be an assembly of machinery to assess	Manufacturer, if any	Does not apply to the design; to assess for any builder who sells
G8	ATEX Directive 2014/34/EU and IEC 60079 series, Equipment for explosive atmospheres	European Union; international	Equipment in atmospheres with flammable gas or dust	Battery drills, the blower and the lamps stay at the portal or are battery devices; whether any item may go where methane or dust is present is a question for the operator's gas assessment	Operator	To assess; not claimed
G9	Indian mines and disaster-response guidance for tunnel rescue (NDMA, DGMS)	India	Rescue practice, equipment holdings and mine safety in the first co-design country	The first co-design candidate is India's National Disaster Response Force (SVD-DDR-001 item 11, not agreed); which guidance governs a tunnel rescue kit to confirm with them	Co-design partner	To assess
G10	CERN-OHL-S-2.0 and MIT licences	International	Terms for the open hardware and the scripts	The repo's licences; the kit's liability disclaimers are in the safety notes	Maintainer	Applies

Status: to assess, applies, does not apply, met by design, certification needed later.

Notes

- The only figures taken from the standards are the 15:1 rope factor (G6) and the existence of the confined-space duties (G1, G2). Section numbers, limits and classifications that were not read at source are not quoted; they are marked to assess.
- Standards shared across the tunnel-rescue focus area are researched once; HatchSide (air and gas practice, SVD-DDR-001 item 12) is the reference and is not edited here.

Sources

- OSHA, "Permit Required Confined Spaces, Notice of proposed rulemaking" (59 FR 60735, Docket S-019A), https://www.osha.gov/node/16671, accessed 2026-11-02. Establishes 29 CFR 1910.146 and its rescue paragraph (k); the final text of the standard is to be read at source.
- FireRescue1, "NFPA 1983: Where did it come from and what does it mean?", https://www.firerescue1.com/fire-products/technical-rescue/nfpa-1983-where-did-it-come-from-and-what-does-it-mean, accessed 2026-11-02 (15:1 rope factor, consolidation into NFPA 2500 on 2021-09-15).
- European Agency for Safety and Health at Work, "Regulation (EU) 2023/1230 on machinery", https://osha.europa.eu/en/legislation/directive/regulation-20231230eu-machinery, accessed 2026-11-02 (applies from 2027-01-20; replaces Directive 2006/42/EC).
- Estonian Centre for Standardisation and Accreditation, EVS-EN ISO 12100:2010 catalogue entry, https://www.evs.ee/en/evs-en-iso-12100-2010, accessed 2026-11-02.
- Hazchem Safety, "BW Gas Alert Micro Clip XL 4 Gas Detector (O2, LEL, CO, H2S)", https://www.hazchemsafety.com/product/bw-gas-alert-micro-clip-xl-4-gas-detector-o2-lel-co-h2s/, accessed 2026-11-02 (GBP 776.52 excluding VAT).
- NFPA 1670, 29 CFR 1926 Subpart AA, ATEX 2014/34/EU, IEC 60079 and the Indian guidance (G2, G3, G8, G9) are named from general knowledge and not read at source in this session: to assess.